

# Integration of Vital Signs Monitoring System Based on Single Antenna Bio-Sensing (SABioS) Method

Mehran Ahadi, Amine Miled, Marc-André Dugas, and Younès Messaddeq

**Abstract**—Accurate respiratory monitoring is essential in critical care settings where abnormal breathing can indicate rapid deterioration and require immediate intervention. Existing non-invasive approaches face trade-offs in comfort, reliability, and portability, limiting their clinical applicability. Single Antenna Bio-Sensing (SABioS) offers an alternative by capturing local dielectric variations with a single wearable antenna, enabling comfortable, non-invasive monitoring of respiratory and cardiac activity. This work presents an implementation of SABioS on a compact custom PCB operating at 915 MHz in combination with a flexible sinusoidal dipole antenna sensor and a dedicated acquisition software, demonstrating the feasibility of portable, low-power, continuous monitoring. The system was evaluated against synchronized medical-grade reference devices, providing the first validation of SABioS for accurate cardiac activity detection in addition to respiration, with respiratory rate agreement of  $\text{MoD} \pm \text{LoA} = 0.005 \pm 0.217$  bpm and heart rate agreement of  $\text{MoD} \pm \text{LoA} = 0.0096 \pm 0.322$  bpm. Additional experiments confirmed stable operation of the system in long-term monitoring and robust performance across wearing configurations and body positions, including recumbent scenarios for sleep monitoring with up to 13.7 dB SNR improvement. Compared to the earlier proof-of-concept setups of SABioS, the integrated system achieves considerably higher signal-to-noise ratio and measurement accuracy, with up to +24.2 dB SNR improvement, underscoring its potential for continuous vital signs monitoring in practical healthcare applications.

**Index Terms**—Respiratory, Cardiac, Vital signs, Respiratory waveform, Biosensing, Antenna sensor, RF sensor, Non-invasive, Biomedical signal processing, Wearable

## I. INTRODUCTION

**R**ESPIRATORY activity is a key vital sign in clinical medicine, widely recognized as an early predictor of serious outcomes such as respiratory failure and cardiac arrest [1], [2]. In intensive care units (ICU), continuous monitoring is critical to assess patient progression and to detect early signs of deterioration post-extubation [3]–[5]. The same need extends

to post-operative settings and step-down wards, where abnormal breathing patterns may indicate complications such as opioid-induced respiratory depression or sepsis [5], [6]. Rapid identification of respiratory pattern changes is essential, since alterations in respiratory activity often signal life-threatening events that must be addressed without delay.

In this context, accurate, comfortable, and non-invasive monitoring remains fundamental to effective healthcare delivery, reducing complications and easing care delivery. This need is especially pronounced for fragile patients in critical situations, or for neonatology and pediatrics patients, where reliable continuous measurement is vital but conventional sensors are difficult to apply and stabilize without compromising comfort or accuracy, potentially interfering with care delivery and causing additional complications [7].

Several non-invasive technologies have been developed to monitor respiratory activity, but each comes with trade-offs. Belt-based methods such as inductive [8] or resistive plethysmography [9] provide continuous tracking of chest expansion, yet they are uncomfortable, interfere with natural breathing, and lose accuracy when displaced [10]. Notably, a multicenter NICU study involving over 8000 neonates reported that 15.55% of iatrogenic skin injuries were medical device-related pressure injuries, underlining the risks associated with such devices in fragile patients [11]. There are also methods that derive respiration indirectly from physiological signals, such as photoplethysmography (PPG) derived, and electrocardiography (ECG) derived respiration [12], [13]. While attractive for their convenience, they rely on extensive signal processing [14] and are highly sensitive to motion artifacts and skin properties, which makes them unreliable in dynamic clinical environments [15]–[17].

RF-based approaches have also been explored, such as radar systems that monitor chest wall motion without contact [18]. While conceptually attractive, radar solutions often require bulky hardware, are vulnerable to multipath interference, and show reduced accuracy in environments with multiple moving subjects [19]. These factors limit their practicality in ICUs, where personnel and visitors are frequently near the patient and can introduce signal disturbances. In addition, radar systems are restricted to a limited field of view, meaning that monitoring is interrupted whenever the patient leaves the bed for procedures or routine movements.

Other RF methods, such as RFID- and Bluetooth RSSI-based techniques [20]–[22], estimate respiratory activity from fluctuations in signal strength [23] or phase shift [24], caused either by stretching of the tag or by chest wall displacement

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that changes the distance between the antenna and receiver. These methods also depend on external readers, are highly sensitive to environmental conditions and signal path variations, and remain constrained by the inherent limitations of third-party communication chips, which reduce signal acquisition quality and reliability. Collectively, these shortcomings underscore the need for a solution that enables real-time, continuous, comfortable, and portable monitoring without compromising accuracy or reliability.

Single Antenna Bio-Sensing (SABioS) offers a novel approach to respiratory monitoring by employing a single wearable antenna sensor that locally detects changes in the body's dielectric properties induced by respiration [25]. When positioned near the chest, the thoracic cavity acts as a variable dielectric medium, altering the antenna's impedance and resonance characteristics and thereby enabling accurate capture of vital signs. Compared to strain-based belts and patches, SABioS eliminates the need for tight-fitting belts, direct skin contact, or strong adhesives, enhancing patient comfort and reducing care complexities. In contrast to radar-, RFID-, or RSSI-based systems, it requires no external readers or measurement stations, which makes its sensing mechanism independent of signal propagation and potentially allows for an implementation in a portable format for continuous real-time monitoring.

It has previously been demonstrated that the SABioS method can accurately detect respiratory activity (RA) through a proof-of-concept (PoC) setup involving a Vector Network Analyzer (VNA) device, validated against both a hospital-grade spirometer [25] and a high-precision optical tracking system [26]. A more recent study using the same setup further demonstrated the versatility of SABioS, showing reliable operation under varying conditions, including different antenna wearing configurations and recumbent positions [27].

However, it remains an open question whether SABioS can be realized in a compact, standalone, and portable system without compromising accuracy or reliability, which is essential for practical deployment of this monitoring technique. Furthermore, while the cardiac activity (CA) signal was previously observed in the antenna response, validation against a medical reference has not been possible due to the technical limitations of the earlier VNA-based PoC setup and also the time-sensitive nature of cardiac signals.

In this work, we present a compact, integrated Printed Circuit Board (PCB) implementation of SABioS that consolidates the core functionality of the system. Section II describes the hardware design, covering the RF circuitry for respiration signal detection as well as the data acquisition and processing components. Section III presents experimental results and recording conditions, while Section IV reports agreement analyses for RA and CA signals alongside stability and robustness tests under different operating configurations. Finally, Section V concludes with a summary of the findings and prospective directions.

## II. THE INTEGRATED SYSTEM DESIGN

The SABioS method allows for capturing vital signs using a single antenna placed near the thorax through two key

mechanisms [25]. The first is the Permittivity Shift Effect (PSE), caused by variations in the dielectric properties of the lungs and surrounding tissues during breathing, which in turn change the distribution of the antenna's near fields. The second is the Antenna Deformation Effect (ADE), which occurs when a flexible antenna becomes detuned due to structural deformation caused by thoracic or abdominal expansion, which correlates with tidal volume, reflecting the different phases of the respiratory cycle [28]. Both PSE and ADE modify the antenna's impedance ( $Z_{\text{ant}}$ ) during respiratory activity. This change in impedance, in turn, alters the antenna's reflection coefficient ( $S_{11}$ ) due to the mismatch with the characteristic impedance of the system ( $Z_0$ ), as shown in Equation (1).

$$S_{11} = \frac{Z_{\text{ant}} - Z_0}{Z_{\text{ant}} + Z_0} \quad (1)$$

Changes in  $S_{11}$  during respiration can be tracked in two ways. The first approach is to monitor the changes in the resonance frequency of the antenna over time by repeatedly scanning a specific frequency range, and for each scan, identifying the frequency at which the minimum  $|S_{11}|$  or the zero crossing of  $\angle S_{11}$  occurs, as one sample point for the signal. Implementing this approach essentially requires a sweeping frequency generator and a wideband RF chain capable of supporting the entire scanned range with a good impedance match, which makes the setup similar in complexity and operation to a full-fledged VNA. It also demands a relatively high-speed ADC and considerable memory in the microcontroller to process the data. These requirements increase circuit complexity, makes miniaturization more difficult, and leads to higher power consumption and fabrication costs. Furthermore, operating in a wide frequency range requires broader frequency band allocation, making this method less practical from a regulatory perspective for real-life applications.

The second and more practical approach, which is the method implemented in this work, is to interrogate the reflection coefficient of the antenna at a fixed interrogation frequency over time, acquiring either the full complex value of  $S_{11}$  or solely its magnitude ( $|S_{11}|$ ) or phase ( $\angle S_{11}$ ). In contrast to the spectral scanning method, this technique eliminates the need for broadband hardware and the associated processing overhead. The core of this design would be a circuit that measures the reflection coefficient of an antenna excited by a single-tone continuous-wave signal generator, and routes the detected signal to a data acquisition subcircuit for presentation, storage and analysis.

### A. The Circuit Setup

A compact custom PCB was developed to include the core functionality of SABioS onto a single board. Figure 1a illustrates the block diagram of the implemented PCB together with the experimental setup used to operate the circuit. To simplify the design, signal generation and data acquisition were managed by external modules, denoted by SIG (LMS-152D, Vaunix Technology, Newburyport, MA, US) and DAQ (NI USB-6212, Emerson Electric, St. Louis, MO, US), respectively, allowing full focus on integration of the most crucial components of the system.

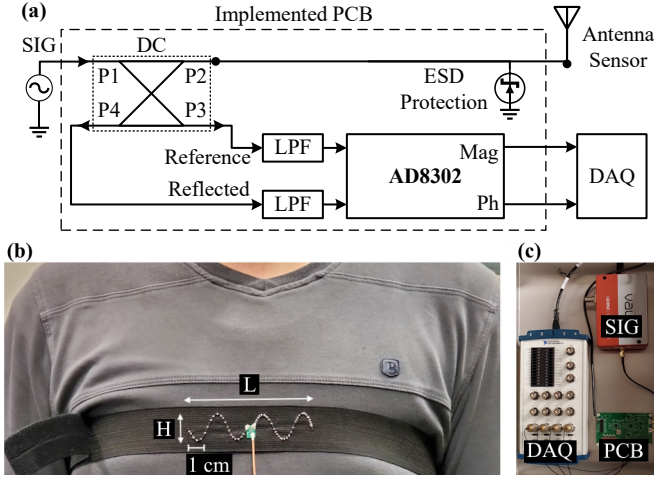


Fig. 1. (a) The setup overview. Dashed rectangle indicates the implemented PCB. (b) The sensor installed over an elastic band worn around the chest (c) The PCB plus the external SIG and DAQ devices

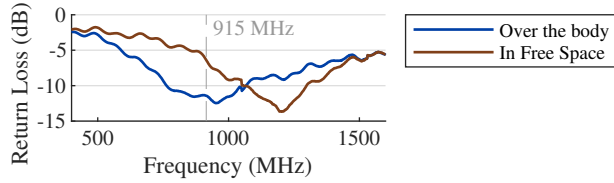


Fig. 2. Measured return loss of the antenna in free space and over the body.

The system is connected to a  $50\Omega$  wearable sinusoidal dipole antenna fabricated from custom flexible conductive polymers consisting of poly(ethylene-co-vinyl acetate) (PEVA) and multi-walled carbon nanotubes (MWCNTs) [29], as shown in Figure 1b. The antenna's geometrical parameters and performance are summarized in Table I and Figure 2. A close proximity to the dense dielectric structure of the body will cause the antenna's frequency response to change [25]. Nevertheless, when placed over the body, the designed antenna maintains a good match with a return loss below  $-10$  dB across a wide range centered around 915 MHz, overlapping with the license-free ISM band in North America [30]. The wide bandwidth of the antenna ensures the PSE can still be captured with the same antenna even when minor frequency shifts occur due to differences in body composition across subjects.

The antenna is fed through a balun located at its feedpoint gap, which converts the unbalanced coaxial signal into a

TABLE I  
ANTENNA GEOMETRICAL PARAMETERS AND MEASURED PERFORMANCE

| Geometrical Parameters       |                       |                            |
|------------------------------|-----------------------|----------------------------|
| Total wire length            | 142 mm                |                            |
| Axial Length (L)             | 80 mm                 |                            |
| Peak-to-peak Height (H)      | 18 mm                 |                            |
| Sinusoidal Peaks per pole    | 3 Peaks               |                            |
| Polymer Wire Thickness       | 1 mm                  |                            |
| Feedpoint Gap Width          | 8 mm                  |                            |
| Performance                  | In Free Space         | Over the Body <sup>†</sup> |
| Best Match Frequency ( $f$ ) | 1190 MHz              | 951 MHz                    |
| Return loss at $f$           | $-13.41$ dB           | $-12.46$ dB                |
| Impedance at $f$             | $51.5 - 20.6j \Omega$ | $40.6 - 19.8j \Omega$      |
| Operating Bandwidth          | [1057, 1299] MHz      | [762, 1067] MHz            |

<sup>†</sup> Male, 29 years, 190 cm, 96 kg, over the chest.

balanced one for proper dipole excitation. The employed balun (TTC1-33W+, Mini-Circuits, Brooklyn, NY) is a wide-band  $50 \Omega$  RF transformer with a typical insertion loss of 1.5 dB, a phase unbalance of  $\pm 4^\circ$ , and an excellent common mode rejection ratio (CMRR) of 26 dB. These characteristics help minimize its influence on the antenna response.

The antenna output pin of the circuit is also equipped with an electrostatic discharge (ESD) protection diode, providing a 9 kV protection, with a negligible capacitance of 0.15 pF, making it ideal for protecting RF lines. For simplicity in the experiments, the antenna was mounted on a flexible band to allow for easy wearing, although it could also be integrated into a T-shirt, skin patch, or necklace, since SABioS benefits from PSE, and does not necessarily require a tight fit or belt for sensing the vital signs [25], [27].

Measurement of the antenna's  $S_{11}$  requires a subcircuit to sample the reflected signal from the antenna. This function can be realized using components such as a directional coupler, a circulator, or a Wilkinson power divider. In this implementation, a directional coupler (DC), with a coupling factor (CF) of 3 dB is employed to extract two signal samples, one sliced from the excitation signal generated by the SIG block and one sliced from the power reflected back from the antenna sensor, which carries the biosensing information.

The S-parameters of an ideal directional coupler ( $S_{DC}$ ) can be written as:

$$S_{DC} = \begin{bmatrix} 0 & \tau & \kappa & 0 \\ \tau & 0 & 0 & \kappa \\ \kappa & 0 & 0 & \tau \\ 0 & \kappa & \tau & 0 \end{bmatrix} \quad (2)$$

where  $\kappa$  is an imaginary number indicating the coupling coefficient and  $\tau$  is a positive real number indicating the transmission coefficient:

$$\kappa = j10^{-\frac{CF}{20}}, \tau = 1 - \kappa\bar{\kappa} \quad (3)$$

Having the  $S_{DC}$ , the power transfer matrix of the directional coupler can also be calculated as  $P_{DC} = S_{DC} \cdot \bar{S}_{DC}$  [31]. Therefore, for an ideal 3 dB directional coupler, we have:

$$S_{DC_{3dB}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & j & 0 \\ 1 & 0 & 0 & j \\ j & 0 & 0 & 1 \\ 0 & j & 1 & 0 \end{bmatrix}, P_{DC_{3dB}} = \frac{1}{2} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad (4)$$

Based on Equation (4), the power generated from the SIG block which is incident on port P1 of the DC block, is split equally between P2, which is connected to the antenna sensor, and P3, which is connected to the detector subcircuit. In the same way, half of the power backscattered from the antenna sensor at P2 will be routed toward P4. The voltage transferred to the Reference and Reflected lines can be written as Equation (5), where  $V_{SIG}$  represents the voltage generated by the SIG block.

$$V_{Reference} = V_{SIG} \cdot \frac{j}{\sqrt{2}} \\ V_{Reflected} = \underbrace{V_{SIG} \cdot \frac{1}{\sqrt{2}}}_{\text{Outgoing P2}} \cdot S_{11} \cdot \frac{j}{\sqrt{2}} \quad (5)$$

Therefore:

$$\frac{V_{Reflected}}{V_{Reference}} = \frac{1}{\sqrt{2}} S_{11} \quad (6)$$

Equation (6) indicates that for an ideal DC, the phase difference between the  $V_{Reference}$  and  $V_{Reflected}$  signals remain

equal to  $\angle S_{11}$ , and the power ratio will be  $\frac{1}{2}|S_{11}|^2$ , showing an approximate loss of 3 dB compared to the actual power reflected by the antenna, since:

$$10 \log_{10} \left( \frac{1}{2} |S_{11}|^2 \right) \approx S_{11, \text{dB}} - 3 \text{ dB} \quad (7)$$

This allows for direct reconstruction of antenna's full complex  $S_{11}$  value using a relative power and phase detector. To preserve the consistency of the signals, the signal traces on the PCB routed from P3 and P4 to the detector subcircuit are kept equal in length to preserve their relative phase and loss.

The component selected for the 3 dB directional coupler is QCS-981+ (Mini-Circuits, Brooklyn, NY), which provides high isolation of 17 dB, minimizing signal leakage between isolated routes of  $P1 \rightarrow P4$  and  $P2 \rightarrow P3$ .

The outgoing signals on P3 and P4 are passed through an analog low-pass filter (LPF) with the cutoff frequency of 1150 MHz to suppress out-of-band harmonics and noise sources. However, unwanted environmental RF signals are not a significant concern since the body's dense and lossy dielectric tissue dominates the antenna's near-field distribution and reduces its far field radiation efficiency [25]. Simulations show an efficiency reduction of about 10 dB near a chest phantom, leaving the captured signal primarily governed by dielectric shifts during respiration. In addition, vital signs appear as sub-Hz envelope variations, whereas commercial communication systems produce baseband components in the kilohertz to megahertz range that are removed during data acquisition and filtering.

The filtered signals are then passed to the AD8302 Power and Phase Detector IC (Analog Devices, Wilmington, MA), which detects input powers down to  $-60$  dBm. The detector outputs are proportional to the relative ratio of its two inputs, with a magnitude output over a  $\pm 30$  dB range at 30 mV/dB and a phase output spanning  $0 - 180^\circ$  at 10 mV/ $^\circ$ . It operates at a voltage as low as 2.7 V with a maximum quiescent current of 27 mA, making it well suited for portable, low-power applications. In the final stage of the circuit, the measured magnitude and phase are passed to the external DAQ block for sampling.

The output power of the SIG block is set to  $-20$  dBm, equalling 0.01 mW. Considering the 3 dB loss of the DC block, and ignoring any signal path losses, the antenna will receive a total power of  $-23$  dBm, and the maximum reflected power arriving at the power detector will be  $-26$  dBm. Since the power detector's minimum input range starts from  $-60$  dBm, the detector will be able to operate with its full dynamic range in measuring antenna's  $|S_{11}|$ , covering from  $-30$  dB to 0 dB. This range reliably covers the return loss of the employed antenna at its resonance, as reported in Table I.

The SIG block is tuned to operate on 915 MHz, which is situated in the license-free 902-928 MHz industrial, scientific, and medical (ISM) band to avoid interfering with regulated bands in the North America region [30]. The Effective Isotropic Radiated Power (EIRP) of the antenna can be calculated by adding the  $+2.51$  dBi gain of the sinusoidal antenna [29], resulting in  $-20.49$  dBm. As calculated in the Supplementary Material Section A, this EIRP is equivalent to a field strength of 5.45 mV/m at 3 meters away [32], which is

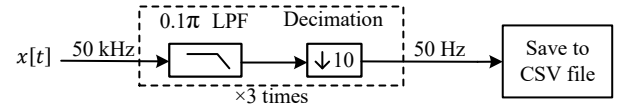


Fig. 3. The real-time downsampling pipeline. Dashed box is repeated three times to obtain the required sampling rate.

well below the 50 mV/m limit defined in FCC Section 15.249 of United States [33] and also the 500 mV/m in ISED RSS-210 Annex F of Canada for ISM band [34]. For Europe region, the antenna can be alternatively designed for the license-free 868-870 MHz range, known as Non-Specific Short Range Devices band, according to ERC/REC 70-03 [35].

As for radio exposure safety, the Specific Absorption Rate (SAR) of the sinusoidal antenna averaged over 1 g of tissue has been reported to be  $< 14$  mW/kg for an excitation power of 0 dBm [29]. In our setup, the power delivered to the antenna feed point is  $-23$  dBm, which corresponds to an estimated SAR of  $< 0.07$  mW/kg. This value is negligible compared with the safety limit of 1.6 W/kg defined by the FCC [36], [37].

### B. Data Acquisition and Real-time Downsampling

The two outputs from the detector subcircuit corresponding to the measured magnitude ( $V_{\text{Mag}}$ ) and phase ( $V_{\text{Ph}}$ ) captured via the DAQ block, which samples the signals at 50 kHz. A software is developed in Python to handle real-time downsampling and storage of the acquired data. This script initializes the SIG block, and starts the DAQ acquisition. It also triggers the acquisition of signals from the reference medical devices simultaneously, ensuring optimal synchronization between the SABioS sensor and the reference measurements.

The recorded data stream is processed through the real-time pipeline illustrated in Figure 3, in which the buffers of 0.2 s are downsampled by a factor of 10 after passing through an anti-aliasing LPF. This downsampling step is repeated three times, resulting in a final sampling rate of 50 Hz. The resulting downsampled signal is saved to a CSV file for further offline processing.

## III. RESULTS

To evaluate the accuracy of the compact PCB, its agreement with medical reference devices, and its performance relative to the earlier VNA-based PoC setup of SABioS, a set of experiments were carried out. For respiratory activity monitoring, a hospital-grade spirometer (Easy on-PC, ndd Medizintechnik AG, Zurich, Switzerland) served as the reference against the RA signal. For cardiac activity monitoring, a finger-clip PPG sensor (CMS50D+, Contec Medical Systems, Qinhuangdao,

TABLE II  
ANTHROPOMETRIC CHARACTERISTICS OF VOLUNTEERS

| # | Gender | Age | Weight (kg) | Height (cm) | BMI (kg/m <sup>2</sup> ) |
|---|--------|-----|-------------|-------------|--------------------------|
| 1 | Male   | 42  | 90          | 180         | 27.8                     |
| 2 | Male   | 58  | 66          | 178         | 20.8                     |
| 3 | Male   | 29  | 96          | 190         | 26.6                     |
| 4 | Male   | 51  | 94          | 180         | 29.0                     |
| 5 | Female | 21  | 58          | 160         | 22.7                     |
| 6 | Female | 33  | 69          | 170         | 23.9                     |

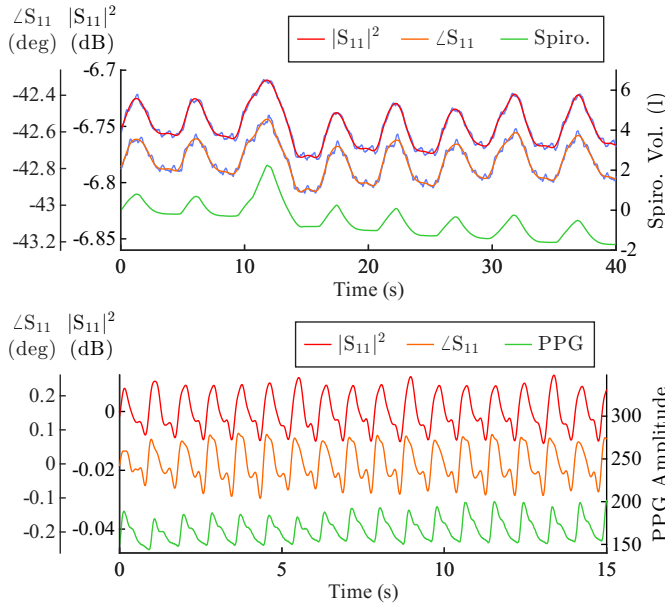


Fig. 4. (Top) Sample of RA signal extracted from magnitude and phase of the  $S_{11}$  against the spirometer reference. The light blue trace under each trace shows the associated raw signal. (Bottom) Sample of CA signal extracted from magnitude and phase of the  $S_{11}$  against the PPG reference. The CA signal has lost the DC baseline due to the applied highpass filter.

China) was used as the reference against the CA signal. The experimental protocol for SABioS has been approved by the Ethics Review Board of CHU de Québec—Université Laval Hospital.

Six volunteers with varied anthropometric characteristics were invited for the experiments to demonstrate the robustness of the system across different body forms and sexes. In the first experiment, the volunteers were asked to wear the antenna sensor operating at 915 MHz and to breathe normally through the spirometer for 120 seconds while sitting. In the second experiment, they wore the antenna sensor and a PPG finger clip simultaneously while sitting for 90 seconds. The anthropometric characteristics of the volunteers for this experiment are presented in Table II.

To evaluate the agreement between the compact PCB and the medical gold standards, an offline processing approach was used. The CSV file containing the downsampled 50 Hz signal was imported into MATLAB, together with the reference signals recorded by the spirometer and the PPG sensor. For analyzing the RA, the signal was filtered using a 0.833 Hz LPF to remove any harmonics above 50 breaths per minute. Similarly, for CA analysis, a bandpass filter designed for 0.833–5 Hz was utilized to only keep the harmonics in 50–300 beats per minute. Both LPFs were FIR type of order 400, designed with a Blackman-Harris window to benefit from its great sidelobe suppression. The acquired RA and CA signals were then aligned with the reference signals to ensure synchronicity and polarity orientation consistency, and segments corrupted by motion artifacts in either the sensor or reference signals were removed.

For the utilized antenna and at the chosen operation frequency close to antenna's resonance, the signals can be generally extracted from both the  $\angle S_{11}$  and the  $|S_{11}|^2$  compo-

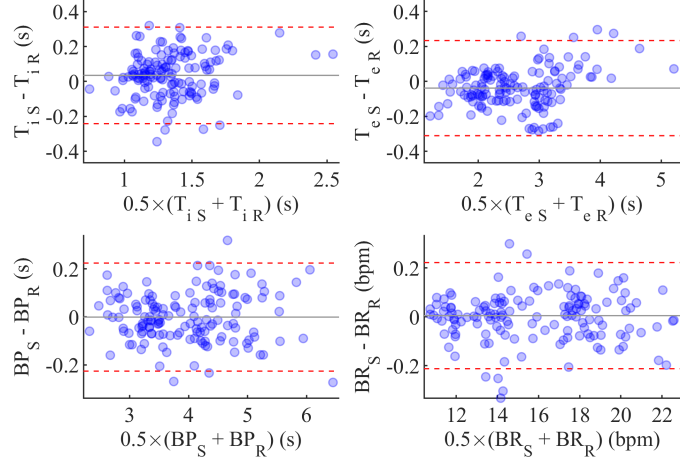


Fig. 5. Bland-Altman plot of respiratory parameters extracted from RA signal. Gray line represents MoD and dashed red lines indicate  $\pm$ LoA.

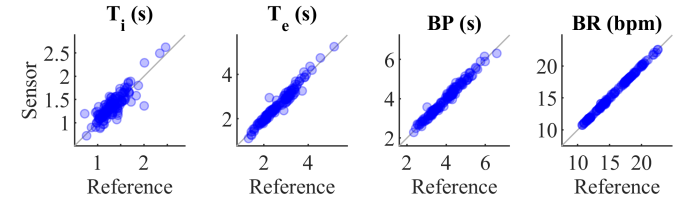


Fig. 6. Scatter plots of the extracted respiratory parameters from RA vs. reference signal, showing the linearity between the sensor and the reference.

TABLE III  
BLAND-ALTMAN  $\text{MoD} \pm \text{LoA}$  OF RESPIRATORY PARAMETERS  
FOR THE COMPACT PCB VERSUS THE EARLIER PoC

| Parameter  | Compact PCB<br>(Current Work) | VNA-based PoC [25]<br>(Earlier Work) |
|------------|-------------------------------|--------------------------------------|
| $T_i$ (s)  | $0.0345 \pm 0.276$            | $0.0051 \pm 0.418$                   |
| $T_e$ (s)  | $-0.0383 \pm 0.272$           | $-0.0020 \pm 0.449$                  |
| $BP$ (s)   | $-0.0011 \pm 0.225$           | $0.0012 \pm 0.401$                   |
| $BR$ (bpm) | $0.0050 \pm 0.217$            | $0.0085 \pm 0.316$                   |

nents. Figure 4 presents a sample recording of the RA signal extracted from the both components of the antenna reflection signal, aligned against the spirometer volume signal as the reference. Similarly, an example of CA signal extracted from the same components is also shown, this time compared with the PPG amplitude signal as the cardiac reference.

## IV. DISCUSSION

### A. Agreement in Respiratory Monitoring

A Bland-Altman analysis was applied to evaluate the agreement between the captured signals and the medical reference measurements. This method is widely used in medical research to assess the agreement between two measurement techniques by quantifying the mean difference (MoD) and the limits of agreement (LoA) [38].

Bland-Altman plots are generated for inspiration time ( $T_i$ ), expiration time ( $T_e$ ), breathing period ( $BP$ ), and breathing rate ( $BR$ ) calculated from the signals recorded across all volunteers. For  $BR$ , a moving average of  $BP$  over five consecutive respiratory cycles were used to calculate the rate. The resulting Bland-Altman plots are shown in Figure 5. Table

TABLE IV  
LINEAR REGRESSION PARAMETERS AND MEASUREMENT ERRORS  
CALCULATED FOR EACH RESPIRATORY PARAMETER

| Parameter | $T_i$   | $T_e$   | $BP$    | $BR$     |
|-----------|---------|---------|---------|----------|
| PB        | 0.9018  | 1.0185  | 0.9883  | 0.9985   |
| $r$       | 0.8645  | 0.9812  | 0.9906  | 0.9994   |
| $p$       | < 0.001 | < 0.001 | < 0.001 | < 0.001  |
| MAE       | 0.11 s  | 0.10 s  | 0.09 s  | 0.08 bpm |
| MAPE      | 8.51 %  | 3.94 %  | 2.23 %  | 0.50 %   |
| MedAE     | 0.09 s  | 0.07 s  | 0.07 s  | 0.06 bpm |
| MedAPE    | 6.47 %  | 3.11 %  | 1.91 %  | 0.37 %   |
| RMSE      | 0.14 s  | 0.14 s  | 0.11 s  | 0.11 bpm |

Defining  $AE(i) = |x_r(i) - x_s(i)|$  and  $APE(i) = |AE(i)/x_r(i)| \times 100\%$ , we have  $MAE = \text{mean}(AE)$ ,  $MAPE = \text{mean}(APE)$ ,  $MedAE = \text{median}(AE)$ ,  $MedAPE = \text{median}(APE)$ , and  $RMSE = \text{sqrt}(\text{mean}(AE^2))$ .

III summarizes the corresponding MoD and LoA values of the current work, and also allows for a side-by-side performance comparison of the current compact PCB versus the earlier VNA-based PoC setup [25]. The smaller LoA of the current work versus the earlier work implies a higher precision in the measurements. Although the calculated MoD is slightly larger in estimating  $T_i$  and  $T_e$ , the value is still negligible and less than 0.04 s across all parameters.

Additionally, proportional bias for each respiratory parameter was also evaluated using the slope of a linear regression between the sensor and the reference values, to show the linearity of the responses, with the corresponding scatter plots illustrated in Figure 6. Table IV lists the proportional bias (PB), Pearson's  $r$ , and the  $p$ -values acquired through the linear regression analysis, along with the calculated measurement errors such as mean absolute error (MAE) and mean absolute percentage error (MAPE). Errors were computed for each observation by comparing the sensor measurements ( $x_s$ ) with the corresponding reference values ( $x_r$ ). To provide a more complete evaluation, the less outlier-sensitive metrics, median absolute error (MedAE) and median absolute percentage error (MedAPE), and also the root mean square error (RMSE) are calculated and presented as well.

The  $BR$  estimation shows a great accuracy across all metrics. Also, the linear regression analysis reveals a great linearity across all parameters, with a slight proportional bias in  $T_i$ . Looking at the calculated absolute errors, the maximum reported is 0.11 s for mean errors, and 0.09 s for median errors, implying a great accuracy between the two measurement methods on average.

### B. Agreement in Cardiac Monitoring

The implementation of SABioS on a custom PCB with its dedicated acquisition software enabled synchronization with external devices, a feature that was technically challenging to achieve with the earlier VNA-based PoC setup. This advancement allowed an agreement study to be performed for the first time between the extracted CA signals using SABioS and the reference PPG measurements.

In this experiment, heart rate ( $HR$ ) was estimated using a moving average of pulse period times over 30 consecutive beats, and the Bland-Altman and linear regression analyses were conducted for all volunteers, with the corresponding plots

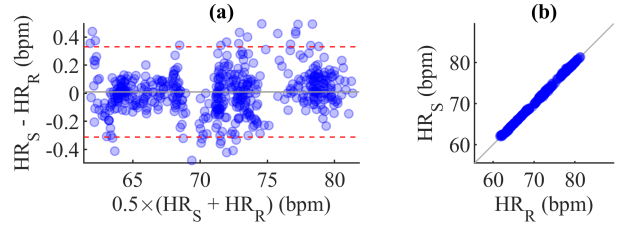


Fig. 7. (a) Bland-Altman plot of HR estimated from the CA signal. The Gray line indicates MoD and dashed red lines represent  $\pm LoA$ . (b) Scatter plots of HR estimated from the CA signal versus the reference signal.

TABLE V  
BLAND-ALTMAN  $MoD \pm LoA$ , LINEAR REGRESSION PARAMETERS,  
AND ERRORS CALCULATED FOR ESTIMATED HR

| Bland-Altman Results      |                        | Calculated Errors |          |
|---------------------------|------------------------|-------------------|----------|
| MoD $\pm LoA$             | 0.0096 $\pm$ 0.322 bpm | MAE               | 0.12 bpm |
| Linear Regression Results |                        | MAPE              | 0.16 %   |
|                           |                        | MedAE             | 0.08 bpm |
|                           |                        | MedAPE            | 0.12 %   |
|                           |                        | RMSE              | 0.16 bpm |
|                           |                        |                   |          |
| PB                        | 1.0048                 |                   |          |
| $r$                       | 0.9996                 |                   |          |
| $p$                       | < 0.001                |                   |          |

shown in Figure 7 and the related parameters summarized in Table V. The results show a great linearity between the HR estimated via the sensor and the reference, with a maximum difference of  $\pm 0.322$  bpm in the 95% percentile.

The obtained results confirm that SABioS can be employed to capture both cardiac activity and respiratory activity using the signal recorded from the same single sensor, and an accurate estimation of heart rate can be obtained.

### C. Reliability in Prolonged Monitoring

Unlike the earlier work with the VNA-based PoC setup, the compact PCB presented in this work supports continuous monitoring, allowing for uninterrupted signal acquisition, which is a key requirement for extended respiratory and cardiac assessments in medical contexts.

To evaluate the stability and reliability of the compact PCB in long runs, extended recordings were conducted with volunteers lying supine on a bed while wearing the antenna sensor for 15 minutes. Figure 8 shows the bandpass-filtered RA signals alongside the time–frequency spectrogram of the raw signal for one of the volunteers, computed using a 40-second Blackman–Harris window with 30-second overlaps. The yellow ridges correspond to the detected RA and CA signals. The moments of deep breathing, shallow breathing, and breathing stop can be seen in the RA signal and the histogram as well.

Obtained results demonstrated consistent tracking of respiratory and cardiac activity throughout the entire session, with no visible instability, deviation, or cutoff in circuit operation, implying that the implemented circuit is stable for continuous RA and CA monitoring.

### D. Monitoring on Different Antenna Locations

An experiment is done to show the performance of the compact PCB while antenna is placed on different regions of the upper body. Figure 9 shows the signals recorded on each

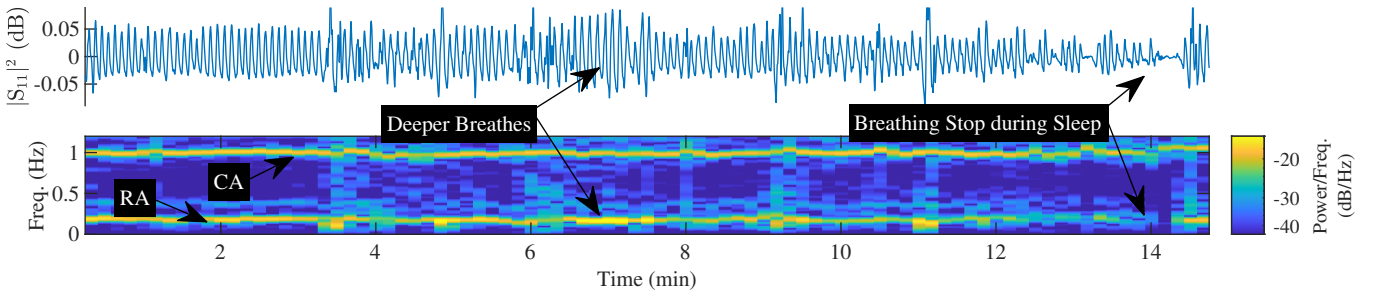


Fig. 8. RA signal and corresponding spectrogram from a long-duration test. Deep-breathing episodes appear as high-intensity regions in the spectrogram. A brief pause in respiration associated with the subject falling asleep is visible near the 14th minute, where the signal flattens and the spectrogram darkens.

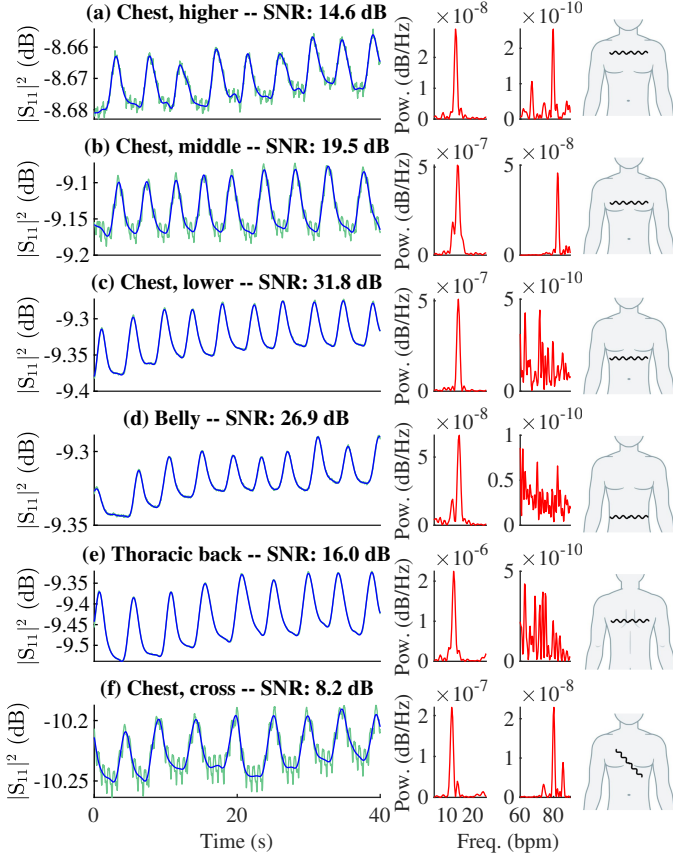


Fig. 9. Recorded signals in different wearing configurations. Blue trace is RA signal and green trace is the raw signal. Two PSDs are shown for each signal, corresponding to RA and CA frequency ranges.

region, along with the Power Spectral Density (PSD) of the signal in respiratory and cardiac frequency ranges.

It is evident that RA signal can be extracted in all of the configurations, but the CA signal weakens as the sensor gets further away from the chest middle area where is the closest to the heart. This observation is in agreement with earlier electromagnetic modelings [25] and experiments [27] on SABioS, confirming that the best location for capturing both RA and CA signals is the chest middle area, whether in horizontal or in crossed configurations, corresponding to subfigures (b) and (f) of Figure 9.

The Signal-to-Noise Ratio (SNR) of the RA signal was calculated for each setting, with the worst case being 8.2 dB, which still indicates strong signal clarity. A similar experiment

TABLE VI  
SNR OF RA SIGNAL CALCULATED FOR CURRENT VS. EARLIER WORK

| Configuration | Compact PCB<br>(Current Work) | VNA-based PoC [27]<br>(Earlier Work) <sup>†</sup> |
|---------------|-------------------------------|---------------------------------------------------|
| Chest, higher | 14.6 dB                       | 5 dB                                              |
| Chest, middle | 19.5 dB                       | 4.7 dB                                            |
| Chest, lower  | 31.8 dB                       | 7.7 dB                                            |
| Belly         | 26.9 dB                       | 2.7 dB                                            |
| Thoracic Back | 16.0 dB                       | -0.1 dB                                           |
| Chest, cross  | 8.2 dB                        | -4.8 dB                                           |

<sup>†</sup> Signal samples and associated PSDs are available for comparison with current work in Supplementary Materials Section B.

had previously been performed in an earlier work using the VNA-based PoC setup [27], and its corresponding SNR values are reported in Table VI together with those obtained in the current work.

The results clearly demonstrate that the compact PCB achieves a substantial improvement in signal quality compared to the earlier work, with increases in SNR of up to +24.2 dB. The representative signals from both the current and earlier work along with their corresponding PSDs are presented side by side in Supplementary Materials Section B for reader's reference.

### E. Monitoring in Recumbent Positions

Signal acquisition in recumbent positions is also investigated. The volunteer is asked to wear the antenna and lie down on a normal household mattress with integrated metallic springs to simulate real-life sleep monitoring capabilities. Signals are recorded in three different poses: supine, left lateral recumbent, and prone. As shown in Figure 10 (a)-(c), both RA and CA frequency ranges show clear PSD peaks, allowing for reliable extraction of both signals using a simple filtering.

In the prone position, the antenna sensor is pressed against the bed and lies in close proximity to the metallic springs of the mattress, which can slightly detune the antenna and reduce its sensitivity to subtle dielectric shifts in body. In such edge cases, higher circuit sensitivity and improved SNR are essential for reliable detection of RA and CA signals. The implemented system successfully meets this requirement, capturing both with clarity, including a well-defined HR signal as shown in Figure 10 (c), compared to the previously-recorded signal from the VNA-based PoC setup, presented in Figure 10 (f).

The superior signal quality achieved using the compact PCB setup allows for a clearer signal extraction in prone

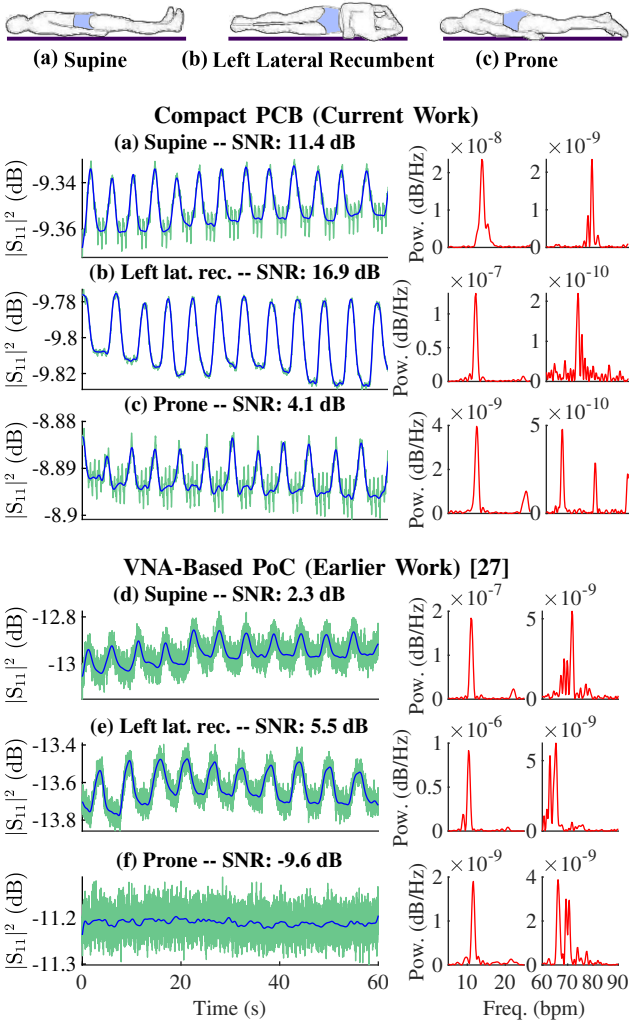


Fig. 10. (a)-(c) Recorded signals in various recumbent positions using the compact PCB introduced in this work. Blue trace is RA signal and green trace is the raw signal. Two PSDs are shown for each signal, corresponding to RA and CA frequency ranges. (d)-(f) Signals of the same experiment previously recorded using the earlier VNA-Based PoC setup [27].

position compared to the earlier VNA-based PoC setup, with an observed SNR increase +13.7 dB for RA signal.

#### F. Comparison with Similar Works

Table VII summarizes reported performance metrics from related work in respiratory monitoring. Compared with these studies, the compact SABioS PCB performs competitively. In particular, our BR MoD $\pm$ LoA of  $0.005\pm 0.217$  bpm compares favorably with the state of the art. Among all entries in the table, the impedance tomography device evaluated by Wisse et al. [39] reports the smallest LoA,  $\pm 0.77$  bpm, yet the reported value is more than three times larger than ours and their method requires installing 32 chest probes on the subject. In contrast, SABioS achieves tighter agreement using a single chest sensor. The remaining works show metrics that are either comparable to SABioS or less favorable. Taken together, these comparisons indicate that integrating SABioS core functionality into a compact PCB based on the introduced circuit design does not degrade accuracy, and that it is a

TABLE VII  
SUMMARY OF SIMILAR WORKS IN RESPIRATORY MONITORING

| Technology                                            | Parameter  | Reported Performance Metrics                                 |
|-------------------------------------------------------|------------|--------------------------------------------------------------|
| Acoustic [40]                                         | $BR$ (bpm) | MoD = 0.63, LoA = $\pm 5.18$                                 |
| Bluetooth RSSI [41]                                   | $T_i$ (s)  | MoD = -0.019, LoA = $\pm 0.511$                              |
|                                                       | $T_e$ (s)  | MoD = 0.032, LoA = $\pm 0.568$                               |
|                                                       | $BP$ (s)   | MoD = 0.003, LoA = $\pm 0.490$                               |
|                                                       | $BR$ (bpm) | MoD = 0.011, LoA = $\pm 2.137$                               |
|                                                       | $BR$ (bpm) | MoD = -0.08, LoA = $\pm 1.47$                                |
| ECG-derived [39]                                      | $BR$ (bpm) | MoD = -0.08, LoA = $\pm 1.47$                                |
| Flexible Fiber Bragg Grating [42]                     | $BR$ (bpm) | MAPE = 1.6%                                                  |
| Flexible Optical Fiber [43]                           | $BR$ (bpm) | MAPE = 5.25%, RMSE = 1.28, MoD = -0.2250, LoA = $\pm 0.8275$ |
| Impedance Tomography <sup>a</sup> [39]                | $BR$ (bpm) | MoD = 0.16, LoA = $\pm 0.77$                                 |
| Nearfield Coherent Sensing [44]                       | $BR$ (bpm) | $r = 0.93$ , MoD = 0.05, LoA = $\pm 5.75$                    |
| Piezoresistive + Inertial Measurement Unit (IMU) [45] | $BR$ (bpm) | MoD = 0.8, LoA = $\pm 2.67$                                  |
| PPG-derived <sup>b</sup> [46]                         | $BR$ (bpm) | MoD = 0.02, LoA = $\pm 2.3$                                  |
| Radar [47]                                            | $BR$ (bpm) | MAE = 0.95, MoD = -0.85, LoA = $\pm 2.18$                    |
| Respiratory Induced Plethysmography [48]              | $BR$ (bpm) | $r = 0.84$ , $p < 0.01$                                      |
| RFID <sup>c</sup> [24]                                | $BR$ (bpm) | 100% - MAPE = 94.21%                                         |

<sup>a</sup> With 32 electrodes installed around the chest <sup>b</sup> Chest patch

<sup>c</sup> Tag over the chest, reader placed at 120 cm

promising solution for continuous, accurate respiratory and cardiac monitoring without compromising comfort.

We also note that most studies report performance metrics primarily for  $BR$ , while quantitative metrics for  $T_i$ ,  $T_e$ , and  $BP$  are rarely included. One contributing factor is that estimating these timing parameters requires a clear, morphology-preserving respiratory waveform, which may not be available across all sensing modalities. A further consideration is that some approaches provide only indirect proxies of respiration or time-averaged outputs, which do not consistently align with the underlying respiratory phases or airflow, thereby limiting the feasibility of deriving timing measures. Examples include ECG- and PPG-derived respiration, which are indirect physiological signals often reported as averages, as well as accelerometer-, radar-, and imaging-based approaches, where waveform shape is strongly influenced by posture and environmental conditions. The clear waveforms obtained via SABioS suggest that this method could potentially enable additional assessments, for example estimating the inspiration to expiration ratio, quantifying breath-to-breath variability, and screening for irregular breathing during sleep or recovery. These possibilities should be explored in future work using dedicated analyses and larger cohorts to further understand the capabilities of SABioS in cardiorespiratory monitoring.

#### V. CONCLUSION

This work presented the first integrated implementation of the Single Antenna Bio-Sensing (SABioS) method on a custom compact PCB operating at 915 MHz with a dedicated acquisition software for continuous data acquisition and processing. It demonstrated the feasibility of embedding SABioS core circuitry in a compact PCB while achieving a higher signal-to-noise ratio and preserving agreement with medical-grade reference devices.

For the first time, this study established the agreement of cardiac signals captured using SABioS with a reference PPG sensor, confirming the accuracy of SABioS in detecting cardiac frequency alongside respiratory monitoring with the same single sensor. This advancement was enabled by the compact PCB implementation, which provided the synchronization and continuous operation capabilities that earlier proof-of-concept setups could not support. This implementation also made possible extended monitoring sessions, which confirmed the stability and reliability of SABioS over long durations without signal degradation.

The performance of the compact PCB was evaluated against medical-grade respiratory and cardiac references, and demonstrated strong agreement with low error margins and great linearity across all tested parameters, with respiratory rate agreement of  $\text{MoD} \pm \text{LoA} = 0.005 \pm 0.217$  bpm and heart rate agreement of  $\text{MoD} \pm \text{LoA} = 0.0096 \pm 0.322$  bpm. Robustness was also validated under different wearing conditions and across multiple recumbent configurations, highlighting its suitability for sleep monitoring and other long-term practical scenarios, while offering an overall SNR improvement of up to 24.2 dB compared with the earlier proof-of-concept setup and enabling reliable prone-position sleep monitoring on a mattress containing internal metal springs with a 13.7 dB SNR improvement. However, establishing medical reliability and broader clinical applicability will require validation on larger and more diverse cohorts, including patient populations, which constitutes an important direction for future work.

Taken together, the accuracy, comfort, portability, and continuous operation of SABioS suggest a promising combination compared with current solutions. That said, antenna placement remains an important factor, with the thoracic region providing the strongest signal quality. Also, the flexible sinusoidal dipole antenna used here can be affected by nearby metallic structures, which may introduce detuning or reduce SNR in situations such as lying on a mattress incorporating metallic springs, wearing clothing with substantial metal components, or holding large metal items very close to the chest. These limitations should be considered when an omnidirectional dipole is used. An alternative antenna design that reduce free space coupling may help mitigate them in future work.

Demonstrating the feasibility of PCB-level integration of SABioS opens the way toward even more compact and autonomous implementations with broader applicability in healthcare and personal monitoring. However, the reliance on external signal generation and data acquisition modules limited full portability for testing across all intended use cases. A fully integrated portable design will enable evaluation of SABioS in realistic mobile conditions such as walking, running, and routine daily activities and will allow assessment of its performance in these practical scenarios, while also paving the way for the development and validation of artifact detection algorithms tailored to common motion conditions. Future work will focus on further integration of hardware and validation across larger and more diverse volunteer populations to expand the robustness and scalability of this method.

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